Positivity of mirror Markov polynomials

Principia Math In Progress

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MathDB 373161. Task 1: a proof of Conjecture 6.2 of [1]. For the mirror deformation defined by the initial segment and signed mutations of [1, Definitions 4.1–4.2], we prove the following, including the absence of internal zero coefficients.

Theorem 1. *Every mirror Markov polynomial $m(q)$ has a strictly positive integer coefficient at each exponent $0, 1, \dots, \deg m$.*

The input from surface cluster algebras is the established matching expansion of Musiker–Schiffler–Williams (MSW), together with coefficient specialization. We give explicitly the cluster realization needed for this application. The positivity theorem for the *squared* deformation in [1] alone does not imply the result for its factors.

1. A paired cluster realization

Use six mutable indices (i, s), where $i \in \{1, 2, 3\}$ and $s \in \{+1, -1\}$. Let B have zero diagonal 2×2 blocks and all-one blocks in positions $(1, 2), (2, 3), (3, 1)$, with the opposite blocks their negatives. Adjoin one frozen variable q and the bottom row

$$c_{i,s} = -s. \tag{1}$$

Initially specialize all six mutable variables to 1. Ordinary geometric-type cluster mutation at an index v uses

$$x_v x_v' = q^{[c_v]_+} \prod_{B_{uv}>0} x_u^{B_{uv}} + q^{[-c_v]_+} \prod_{B_{uv}<0} x_u^{-B_{uv}}, \qquad [a]_+ = \max(a, 0). \tag{2}$$

The extended matrix mutates by the usual matrix-mutation formula, as recalled in [2, Section 2]. Define a *paired step at i* to mutate both $(i, +1), (i, -1)$ and then interchange their labels. These two mutations commute. A direct block calculation shows that each paired step sends B to $-B$: the two intermediate paths add two arrows against each arrow between the other two pairs, reversing it after cancellation.

Here is the complete normalization calculation. At any paired seed let j be the other pair with $B_{(j,t),(i,s)} = 1$, and let k be the remaining pair. There are integers m_i and half-integers $u_{i,s}$ such that

$$c_{i,s} = m_i - s, \quad z_{i,s} = q^{u_{i,s}} x_{i,s}, \quad D_i = u_{i,+1} + u_{i,-1}, \quad m_i = D_j - D_k. \tag{3}$$

Initially $m_i = u_{i,s} = D_i = 0$. The last equation is imposed at each pair with its own positive-column neighbor. Put

$$a = [m_i + 1]_+ + [m_i - 1]_+, \qquad b = [-m_i - 1]_+ + [-m_i + 1]_+; \qquad a - b = 2m_i.$$

Matrix mutation and the label interchange give

$$m_i' = -m_i, \qquad m_j' = m_j - b, \qquad m_k' = m_k + a. \tag{4}$$

Before interchanging labels define

$$\widetilde{u}_{i,s} = D_j - [m_i - s]_+ - \frac{s}{2} - u_{i,s}, \qquad u'_{i,s} = \widetilde{u}_{i,-s}, \tag{5}$$

and keep the other u's unchanged. Then $D_i' = 2D_j - a - D_i$. Since the block orientation reverses, the three required new identities in (3) follow respectively from

$$D_k - D_j = -m_i, \quad D_i' - D_k = m_j - b, \quad D_j - D_i' = m_k + a.$$

This proves (3)–(5) inductively for *every* sequence of paired steps. No restriction on the size or sign of m_i has been used.

Write $A_i = z_{i,+1}z_{i,-1}$. Substituting (3) and (5) into (2), and using $[v]_+ - [-v]_+ = v$, gives precisely

$$z'_{i,+1} = \frac{q^{1/2}A_j + q^{-1/2}A_k}{z_{i,-1}}, \qquad z'_{i,-1} = \frac{q^{-1/2}A_j + q^{1/2}A_k}{z_{i,+1}}. \tag{6}$$

Initially $z_{i,+1} = z_{i,-1} = 1$. Consequently $z_{i,-1}(q) = z_{i,+1}(q^{-1})$ throughout. Multiplying the two equations in (6) gives

$$A_iA_i' = A_j^2 + (q + q^{-1})A_jA_k + A_k^2. \tag{7}$$

Equation (6), with the cyclic order reversed at every step, is exactly the geometric mirror mutation of [1, Equation (4), Proposition 4.8]. That proposition identifies it, up to powers of $q^{1/2}$, with the signed polynomial mutation on every branch. For example, the first normalized polynomials are $1+q$, $1+2q+2q^2$, $2+4q+5q^2+2q^3$, and $3+8q+10q^2+6q^3+2q^4$, with their reciprocal choices when the initial orientation is reversed. Thus every polynomial in the question is a monomial multiple of one of the specialized cluster variables in (2).

2. The surface and the gap-free matching expansion

Realize the mutable matrix as the signed adjacency matrix of the tetrahedral triangulation of the four-punctured sphere. The opposite edge pairs are

$$(12, 34), \qquad (13, 24), \qquad (14, 23).$$

Choose the orientation giving the block matrix above. Each triangular face contributes one cyclic arrow between the three pairs. Flipping an opposite pair again gives a tetrahedral triangulation: after the first flip there are two different arcs between the opposite endpoints, and flipping the original opposite edge replaces it with an arc between the first endpoints. All four triangles still have three distinct vertices. In particular, all arcs reached by paired steps are ordinary arcs, all tagged plain. No notched-arc matching formula is needed.

Use principal coefficients $y_{i,s}$ temporarily. The MSW expansion [2, Theorem 4.9], evaluated at all initial mutable variables equal to 1, is

$$F_\gamma(y) = \sum_{P \in \mathcal{M}(G_\gamma)} \prod_{T \in H(P)} y_{\ell(T)}. \tag{8}$$

Here G_γ is the ordinary snake graph of the arc, $\mathcal{M}$ its perfect matchings, and $H(P)$ the tiles enclosed by the symmetric difference of P and the fixed minimal boundary matching. The label $\ell(T)$ is the initial arc crossed at that tile. The initial tetrahedral triangulation has no self-folded triangles, so there are no substitutions in these tile labels. For an initial arc the expansion is simply 1.

The coefficient specialization formula [2, Theorem 2.13] identifies our frozen row (1) with $y_{i,s} = q^{-s}$. Its tropical denominator is a single monomial. It follows that every specialized x_γ in (2) is a monomial multiple of

$$\sum_{P\in\mathcal{M}(G_\gamma)} q^{h(P)}, \qquad h(P) = \sum_{T\in H(P)} (-s(\ell(T))). \tag{9}$$

Each tile therefore has height weight $+1$ or -1.

Lemma 2. *If every tile of an ordinary snake graph has weight $+1$ or -1, the matching sum in (9) has a positive integer coefficient at every integer between its smallest and largest exponents.*

Proof. The perfect matchings of a snake graph are connected by square-tile flips. For completeness, take an end tile. It adds two vertices to the preceding snake graph. A matching either contains the edge joining these two new vertices, or contains both edges joining them to the old vertices. In the second case, flip that tile, inserting the joining edge and the shared edge. Now remove the joining edge and the two new vertices. What remains is a perfect matching of the shorter snake. Induction connects all such matchings, and their flips lift while keeping the end joining edge fixed. The one-tile case is immediate.

Under one tile flip the height sets relative to the minimal matching differ by adding or deleting that one tile. This also follows directly by taking symmetric differences with its boundary in the definition of $H(P)$. Consequently $h(P)$ changes by 1 or -1. A path of flips from a matching of minimum height to one of maximum height visits every intermediate integer. Every summand has coefficient 1, which proves the claim. □

3. Completion of Task 1

By (3), (6), (9), and Lemma 2, every mirror polynomial has positive integer coefficients throughout its support interval. The source already proves that these representatives are polynomials [1, Theorem 4.4]. We check that their constant terms do not vanish, to obtain exactly the interval in Theorem 1.

After the initial segment, order a mirror triple (a, b, c) by its classical values. Inductively its largest entry c has strictly largest degree and all constant terms are nonzero. This starts at $(1, 1+q, 1+2q+2q^2)$. The corresponding squared Laurent polynomials have the same degrees. If those degrees are $r \le s < t$, (7) sends a forward mutation to degree $2t-r$ or $2t-s$, strictly larger than t; all numerator coefficients are nonnegative, so the leading terms cannot cancel. This is also the degree argument underlying [1, Proposition 2.3]. A new polynomial is, in the notation of the source,

$$c' = q^{\deg c}\frac{q^{\varepsilon}H + C}{\widetilde{d}}, \qquad \varepsilon \in \{-1, 1\}, \quad H = h(q)h(q^{-1}), \quad C = c(q)c(q^{-1}), \tag{10}$$

where h is the retained smaller entry and d the replaced entry. Because $\deg c > \deg h$, the lowest exponent of $q^{\deg c+\varepsilon}H$ is nonnegative. The other numerator term has a strictly positive constant term, and $\widetilde{d}$ has a strictly positive constant term. Thus the polynomial quotient in (10) has a strictly positive constant term. The initial entries $1, 1, 1+q$ and $1, 1+q, 1+2q+2q^2$ satisfy this explicitly. This closes the induction. The support interval consequently starts at zero and ends at the degree, proving Theorem 1.

Attribution and scope. The squared deformation, signed recursion, polynomiality, and equivalence with (6) are established in [1]. The surface expansion and coefficient-specialization theorems are established in [2]; they are not new claims here. The explicit frozen row, the all-seed normalization (3)–(6), and its application to all mirror branches supply the link needed for the conjecture. Neither the mirror uniqueness conjecture nor the separate branch-constant conjecture is asserted here.

Verification. The accompanying exact verifier checks the tetrahedral matrix, 93 paired mutation paths of lengths at most five, the normalization and reciprocity at every step, the printed initial polynomials, and the tile-flip and gap claims on every north/east snake with at most six tiles and all ± 1 tile weights. These checks support the proof; the finite range is not its justification. Review is by the authoring agent only, without independent peer review or proof-assistant verification.

Proposed follow-up tasks

Task 2: **PROPOSED.** Determine a direct construction of the labeled snake graph of $m_{a/b}(q)$ from the continued fraction of a/b, including its minimum-height normalization.

Task 3: **PROPOSED.** Prove or disprove unimodality of the full coefficient sequence of every mirror Markov polynomial.

Task 4: **PROPOSED.** Derive closed formulas for the constant and leading coefficients from the signed left/right mutation word.

Task 5: **PROPOSED.** Classify integer frozen rows on the tetrahedral quiver for which every paired mutation variable has a gap-free polynomial specialization after a monomial normalization.

References

[1] L. Bittmann, P. Jouteur, E. Kantarcı Oğuz, M. Molander, and E. Yıldırım, *A mirror deformation of Markov numbers*, arXiv:2602.14802v1 (2026).

[2] G. Musiker, R. Schiffler, and L. Williams, *Positivity for cluster algebras from surfaces*, Adv. Math. **227** (2011), 2241–2308, doi:10.1016/j.aim.2011.04.018. Theorem numbering here follows the author-hosted version.